



# TOP 5000 MOVIES

DATA ANALYSIS AND  
LINEAR REGRESSION

 Play

# What's playing?



**The  
Shawshank  
Redemption**



**The  
Godfather**

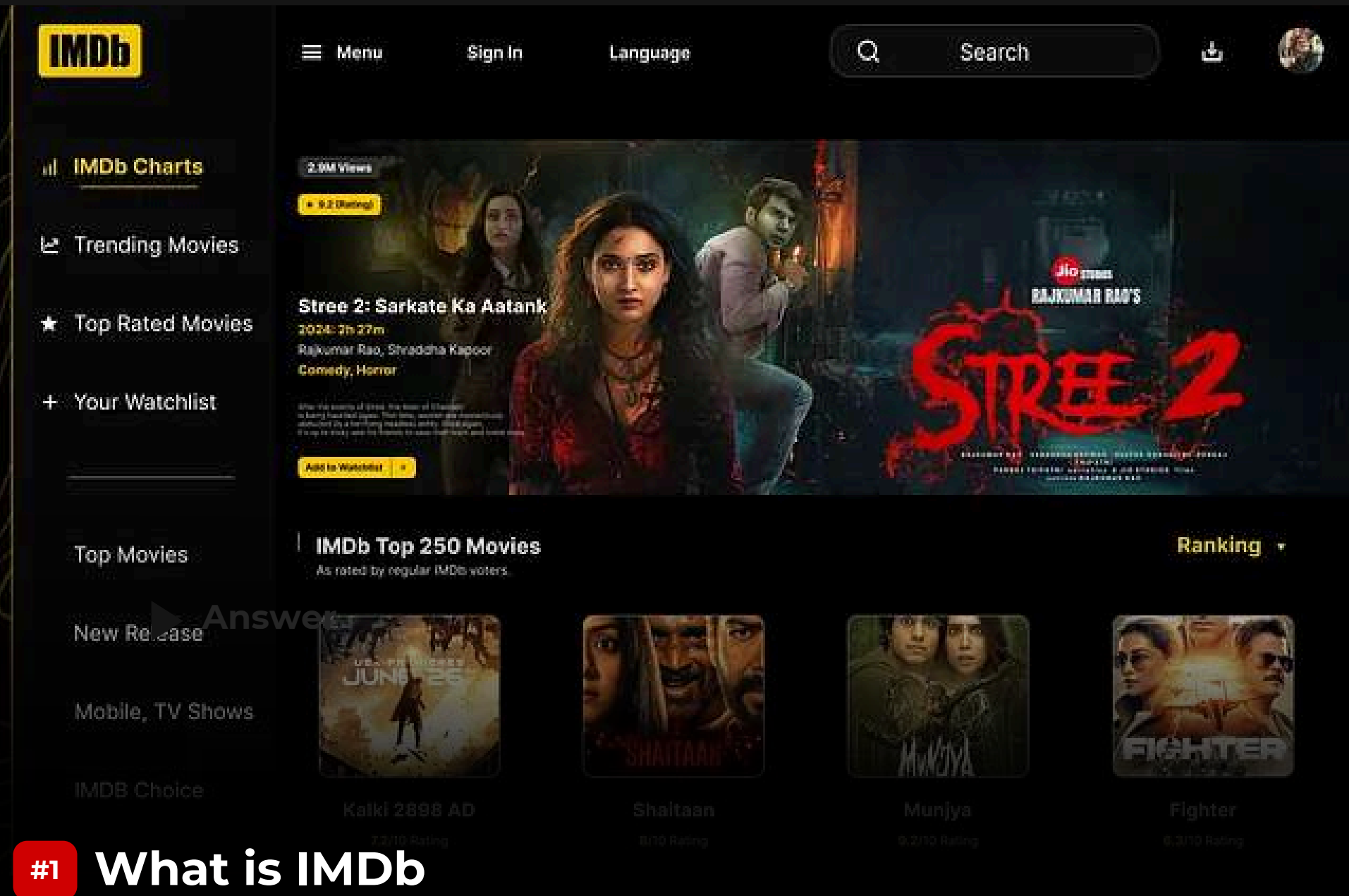


**The Dark  
Knight**



**The Lord of  
the Rings:  
The Return of  
the King**

**Created by:  
Chanoudom Tann  
ID: 6610755487**



The screenshot shows the IMDb website interface. At the top, there's a navigation bar with the IMDb logo, a menu icon, 'Sign In', 'Language', a search bar, and a user profile icon. The left sidebar contains links to 'IMDb Charts', 'Trending Movies', 'Top Rated Movies', and 'Your Watchlist'. The main content area features a large movie banner for 'Stree 2: Sarkate Ka Aatank' with a 9.2 rating and 2.9M views. Below this is the 'IMDb Top 250 Movies' section, which lists four movies: 'Kalki 2898 AD' (7.2/10 rating), 'Shaitaan' (8.0/10 rating), 'Munjya' (8.2/10 rating), and 'Fighter' (8.3/10 rating). The word 'Answer' is overlaid on the left side of the screenshot.

## #1 What is IMDb

- IMDb = Internet Movie Database
- Contains ratings from millions of viewers
- Dataset used: Top 5000 movies



# Dataset Introduction



a. Title Constants

b. Movie Title

c. Release Year

d. Ranking

e. Overall Rating

f. Number of Voters

g. Runtime(in min)

h. Directors

i. Writers

j. Genres

k. IMDb Link

l. Title\_IMDb\_Link

|   | scored   | primaryTitle                                  | startYear | rank | overallRating | numVotes | runtimeMinutes | directors            | writers                                           | genres                    | IMDbLink                                          | Title_IMDb_Link                                   |
|---|----------|-----------------------------------------------|-----------|------|---------------|----------|----------------|----------------------|---------------------------------------------------|---------------------------|---------------------------------------------------|---------------------------------------------------|
| 0 | 10011161 | The Shawshank Redemption                      | 1994      | 1    | 9.3           | 2120165  | 142            | Frank Darabont       | Stephen King, Frank Darabont                      | Drama                     | <a href="https://www.imdb.com/title/tt0111161"... | <a href="https://www.imdb.com/title/tt0111161"... |
| 1 | 10008646 | The Godfather                                 | 1972      | 2    | 9.2           | 2176954  | 125            | Francis Ford Coppola | Mario Puzo, Francis Ford Coppola                  | Crime, Drama              | href="https://www.imdb.com/title/tt0068646"...    | href="https://www.imdb.com/title/tt0068646"...    |
| 2 | 10468568 | The Dark Knight                               | 2008      | 3    | 9.1           | 3095565  | 152            | Christopher Nolan    | Jonathan Nolan, Christopher Nolan, David S. Goyer | Action, Crime, Drama      | href="https://www.imdb.com/title/tt0468568"...    | href="https://www.imdb.com/title/tt0468568"...    |
| 3 | 10167260 | The Lord of the Rings: The Return of the King | 2003      | 4    | 9.0           | 2118280  | 201            | Peter Jackson        | J.R.R. Tolkien, Fran Walsh, Philippa Boyens, P... | Adventure, Drama, Fantasy | href="https://www.imdb.com/title/tt0167260"...    | href="https://www.imdb.com/title/tt0167260"...    |
| 4 | 10108052 | Schindler's List                              | 1993      | 5    | 9.0           | 1555434  | 195            | Steven Spielberg     | Thomas Kennedy, Steven Zaillian                   | Biography, Drama, History | href="https://www.imdb.com/title/tt0108052"...    | href="https://www.imdb.com/title/tt0108052"...    |



# PROJECT GOAL

1. UNDERSTAND WHAT FACTORS INFLUENCE IMDB RATINGS
2. EXPLORE RELATIONSHIPS BETWEEN FEATURES (GENRE, VOTES, RUNTIME, YEAR) AND RATING
3. BUILD A SIMPLE REGRESSION MODEL TO PREDICT THE RATING USING THESE FEATURES
4. COMPARE MODELS (LINEAR, RIDGE, LASSO) USING CROSS-VALIDATION



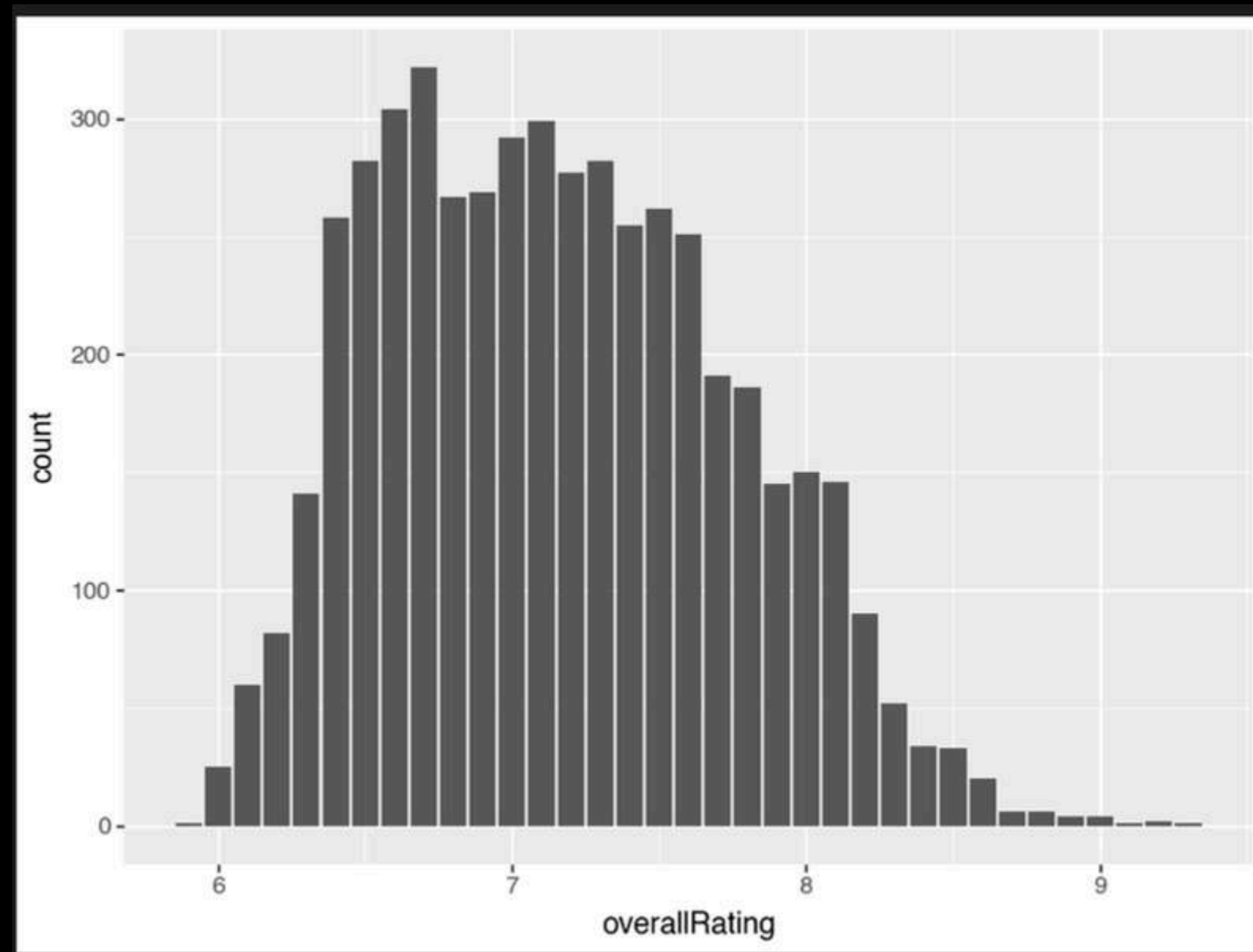
# Feature Engineering

## Extracting main\_genre

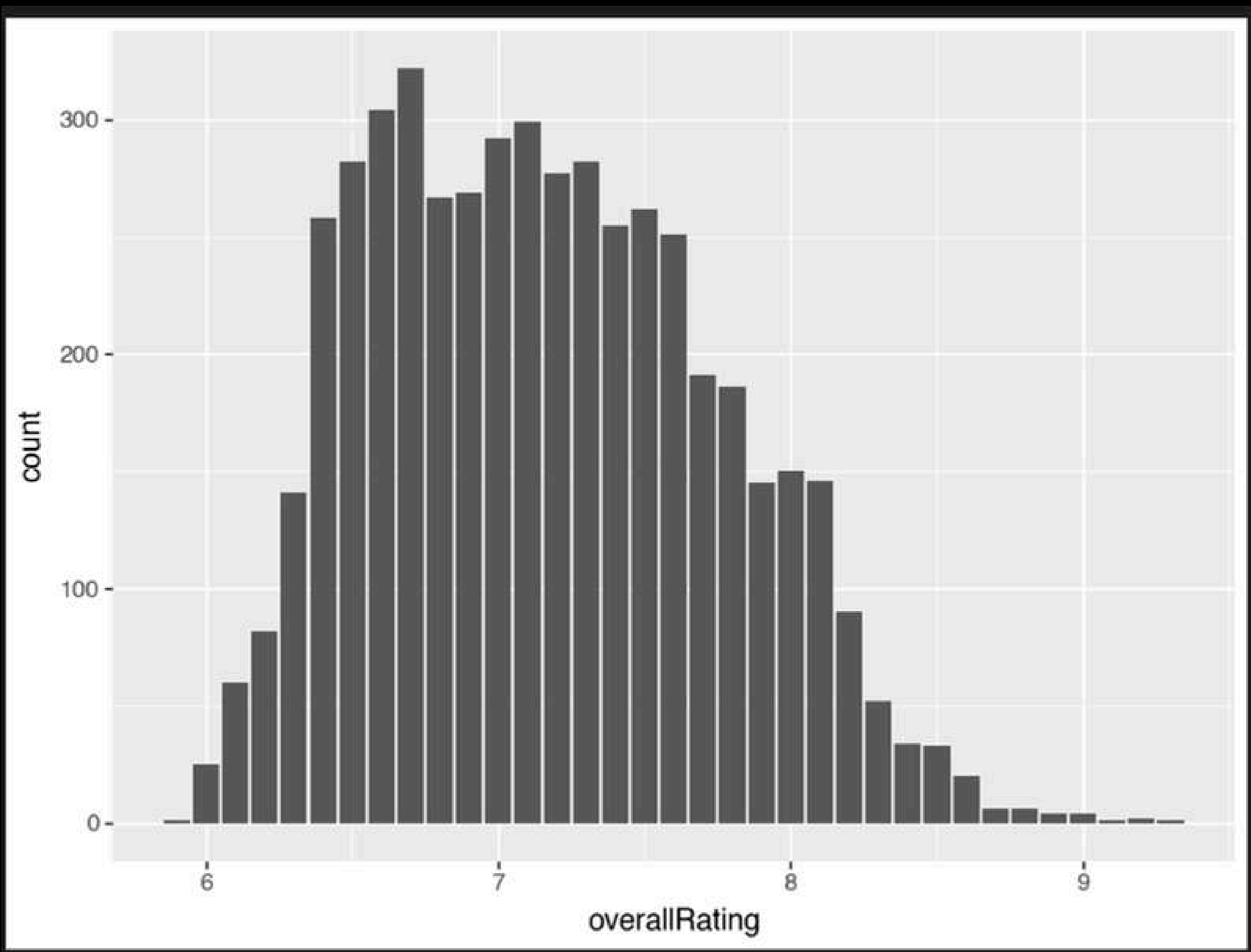
- Movies often have multiple genres (e.g., "Drama, Crime").
- To simplify analysis and improve model performance, the first listed genre is extracted as main\_genre.
- This allows us to treat genre as a single categorical variable in regression.

|   | genres                    | main_genre |
|---|---------------------------|------------|
| 0 | Drama                     | Drama      |
| 1 | Crime, Drama              | Crime      |
| 2 | Action, Crime, Drama      | Action     |
| 3 | Adventure, Drama, Fantasy | Adventure  |
| 4 | Biography, Drama, History | Biography  |

# RATING DISTRIBUTION



- Ratings in this dataset range from 5.9 to 9.3
- Most movies cluster between 6.7 and 7.6
- Important context before evaluating model performance





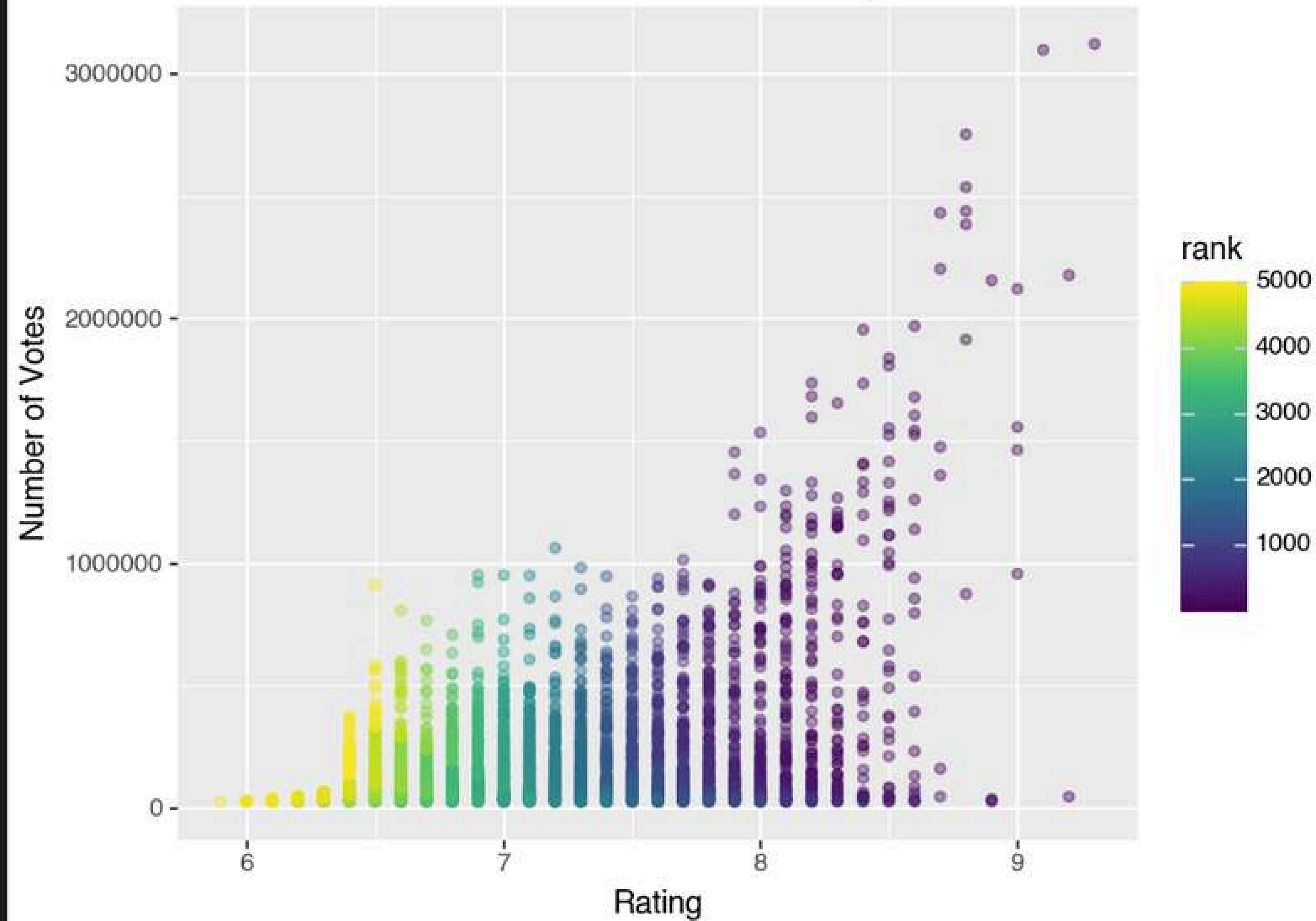
# RELATIONSHIP BETWEEN VOTES AND RATING



- Movies with more votes tend to have slightly higher ratings
- A positive trend is visible
- Some outliers exist with high votes but average ratings
- Each point = one movie.
- Movies with more votes tend to have slightly higher ratings.
- Popular movies (high votes) usually reflect stronger audience consensus.
- Outliers exist: some highly-voted movies still have average ratings.



Number of Votes vs Rating



# RATING DISTRIBUTION BY GENRE



- Shows how movie ratings are distributed across genres
- Action appears the most because it has the largest number of movies in the dataset
- Some genres (e.g., Western) appear less often but tend to have higher ratings
- Highlights both frequency and spread of ratings across categories







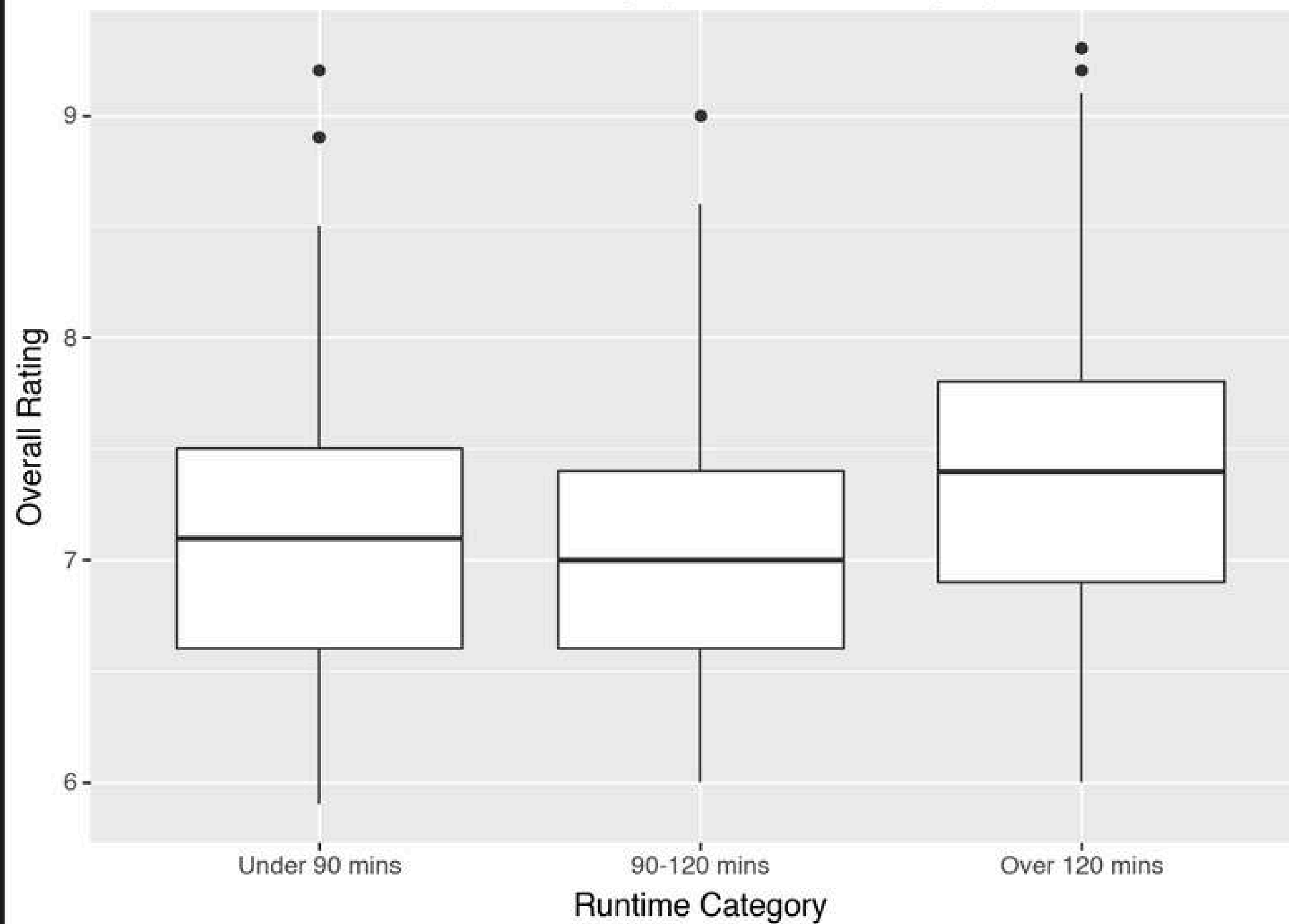
# OVERALL RATING BY RUNTIME



- Movies over 120 mins have the highest median rating.
- Under 90 mins and 90–120 mins show similar medians, both lower than long movies.
- Longer movies show wider rating variation.
- Outliers appear in all categories.



Overall Rating by Runtime Category

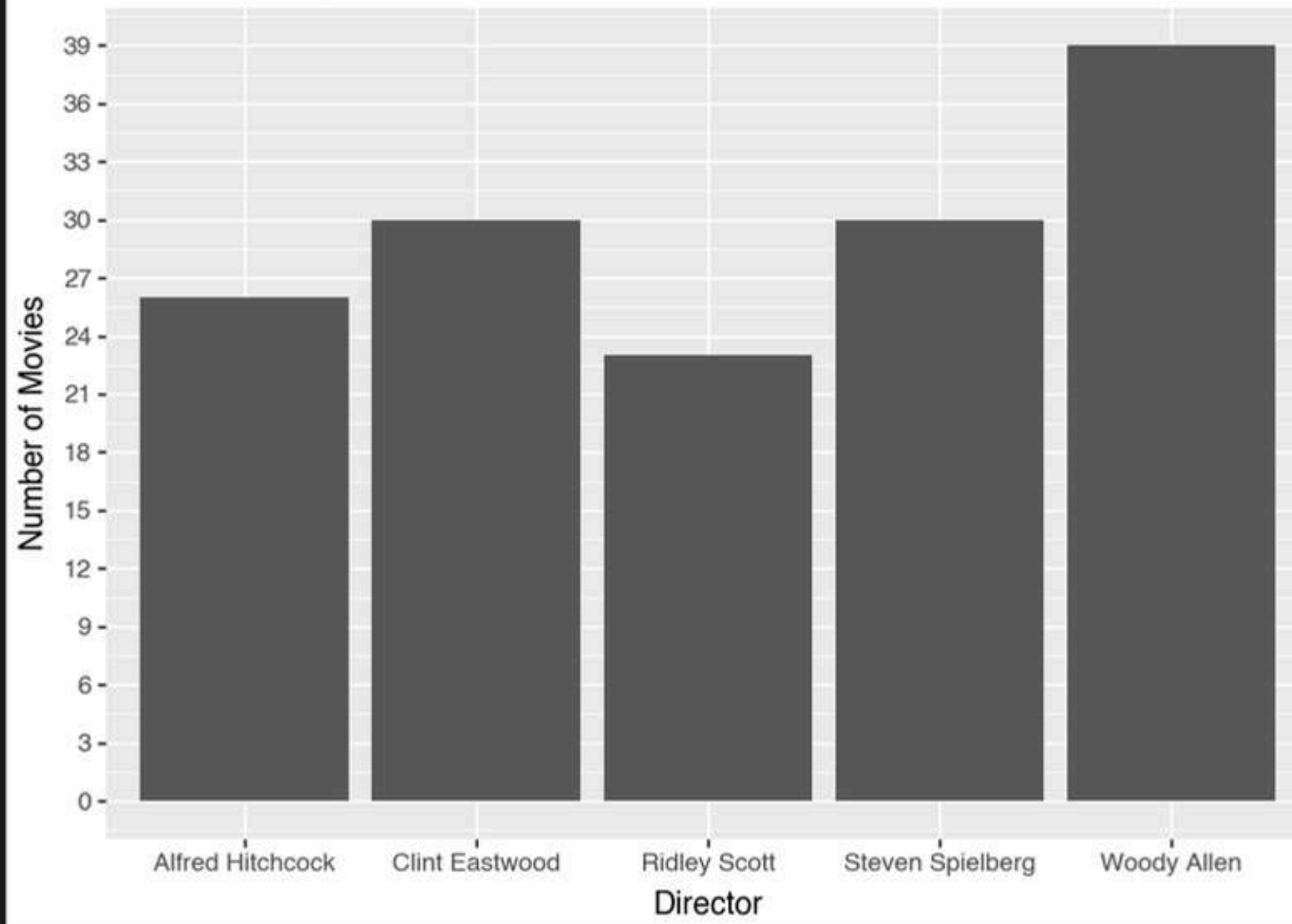


# TOP 5 DIRECTORS

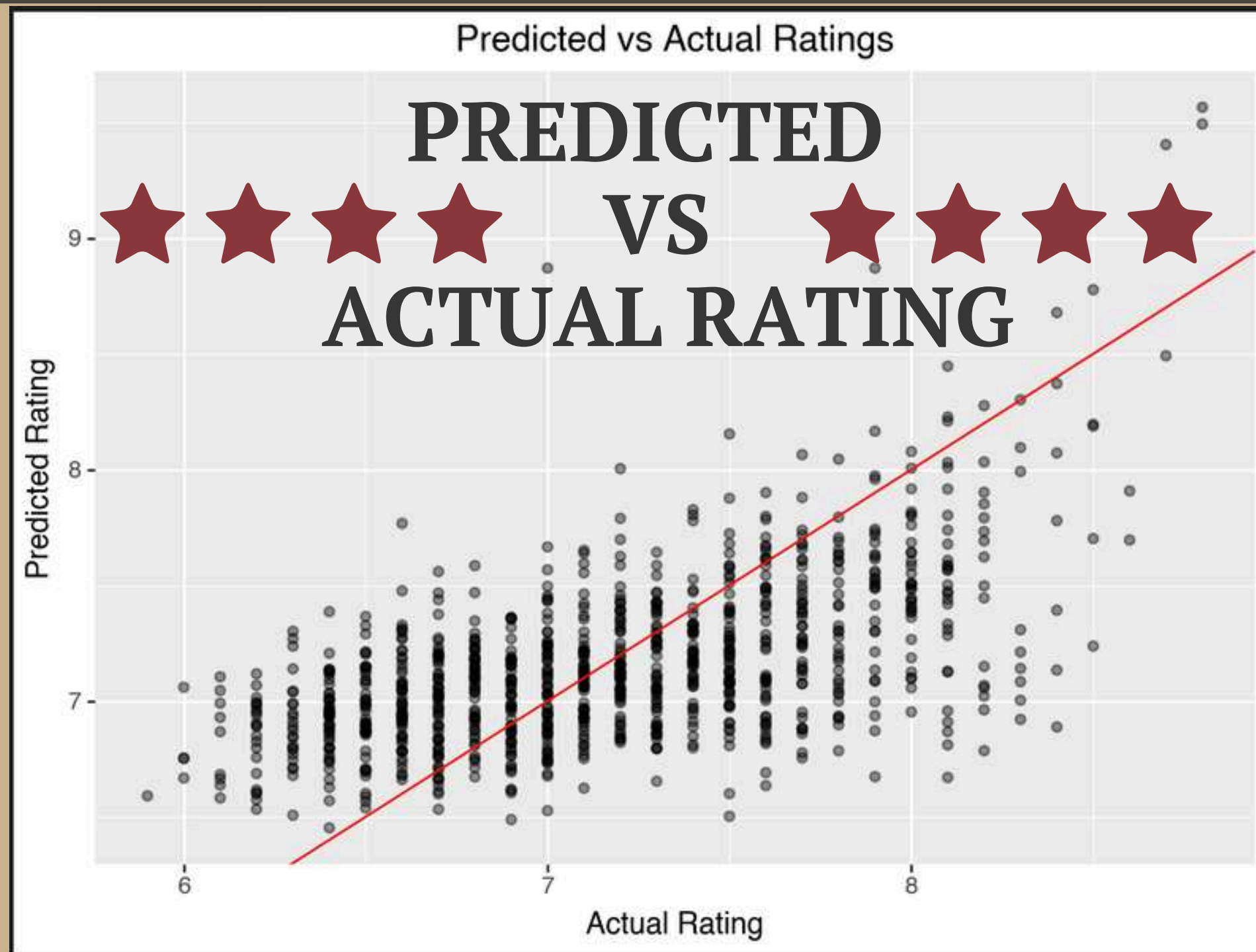
- Shows which directors appear most frequently in the dataset
- Woody Allen has the highest number of movies
- Followed by Clint Eastwood and Steven Spielberg
- Helps us understand movie distribution across directors



Top 5 Directors with the Most Movies in the Dataset



0123456789



0123456789



# IMDB TOP 5000 MOVIES

- Best model: Ridge Regression ( $\alpha = 0.1$ )
- Test MSE: 0.21
- Prediction error(RMSE)  
 $\approx 0.46$  rating points
- Good accuracy given rating range (6–9.3)

|    | Feature Set | Model                         | Mean CV MSE |
|----|-------------|-------------------------------|-------------|
| 0  | base        | Linear Regression             | 0.346923    |
| 1  | movie       | Linear Regression             | 0.246131    |
| 2  | full        | Linear Regression             | 0.244186    |
| 3  | base        | Ridge Regression $\alpha=1$   | 0.346923    |
| 4  | movie       | Ridge Regression $\alpha=1$   | 0.246131    |
| 5  | full        | Ridge Regression $\alpha=1$   | 0.244186    |
| 6  | base        | Ridge Regression $\alpha=0.1$ | 0.346923    |
| 7  | movie       | Ridge Regression $\alpha=0.1$ | 0.246131    |
| 8  | full        | Ridge Regression $\alpha=0.1$ | 0.244186    |
| 9  | base        | Lasso Regression $\alpha=1$   | 0.347840    |
| 10 | movie       | Lasso Regression $\alpha=1$   | 0.347840    |
| 11 | full        | Lasso Regression $\alpha=1$   | 0.347840    |
| 12 | base        | Lasso Regression $\alpha=0.1$ | 0.347840    |
| 13 | movie       | Lasso Regression $\alpha=0.1$ | 0.275353    |
| 14 | full        | Lasso Regression $\alpha=0.1$ | 0.275353    |

▶ Play



- RATINGS RANGE FROM ABOUT 6 TO 9
- GENRE AND VOTES INFLUENCE RATINGS
- WESTERN HAS HIGHEST AVERAGE  
RATING
- RIDGE REGRESSION PERFORMS BEST
- TEST MSE: 0.21 (~0.46 RATING ERROR)

SUMMARY



THANKS FOR

YOUR ATTENTION

▶ End

